

Deep CNN Architectures, Data Strategies, and Regularization Techniques

Table Of Content

1. Deep CNN Architecture and Innovations
2. Large-Scale Dataset and Preprocessing
3. Overfitting Prevention and Regularization
4. Training Methodology and Results

1. Deep CNN Architecture and Innovations

- Our deep CNN has 8 layers: 5 convolutional, 3 fully-connected, ending with a 1000-way softmax.
- Key innovations include ReLU nonlinearities for faster training and overlapping pooling for better generalization.
- Multi-GPU training spreads the model across GPUs, reducing memory constraints and training time.
- Local response normalization creates competition among neurons, aiding generalization and performance.
- Dropout regularization prevents overfitting by randomly dropping neurons during training.

1.1 ReLU Nonlinearity and Training Speed

- ReLUs are non-saturating nonlinearities, $f(x) = \max(0, x)$.
- Deep CNNs with ReLUs train several times faster than those with tanh units.
- ReLU's faster training enables experimenting with larger, deeper models on large datasets.
- ReLUs do not require input normalization to prevent saturation, unlike tanh.

1.2 Overlapping Pooling and Normalization Techniques

- Overlapping pooling ($s < z$) reduces top-1 and top-5 error rates compared to non-overlapping pooling.
- This scheme makes models slightly more resistant to overfitting during training.
- Local response normalization creates competition among neuron outputs, aiding generalization.
- Response normalization applies lateral inhibition, inspired by real neurons, to improve performance.

1.3 Overall Network Structure and Connectivity

- The network contains eight layers with weights: five convolutional and three fully-connected.
- The first five layers are convolutional, with pooling and normalization layers interleaved as specified.
- Response-normalization layers follow the first and second convolutional layers; max-pooling layers follow normalization and the fifth convolutional layer.
- All convolutional and fully-connected layers use ReLU nonlinearities, enhancing training speed and performance.
- Connections are partially GPU-specific: some layers connect only within the same GPU, others across GPUs, optimizing memory use and computation.

1.4 Impact of Depth and Architectural Choices

- Deeper networks consistently outperform shallower ones, demonstrating the importance of depth.
- Removing convolutional layers degrades accuracy by about 2%, highlighting their critical role.
- Experiments show that depth enables learning more complex features, improving recognition.
- Architectural choices like overlapping pooling and local normalization further enhance performance.
- Design decisions are justified by improved training speed, reduced overfitting, and better generalization.

2. Large-Scale Dataset and Preprocessing

- ImageNet contains over 15 million labeled high-resolution images across roughly 22,000 categories.
- For training, images are resized by rescaling shortest side to 256 pixels, then center-cropped to 256×256.
- Images are preprocessed by subtracting the mean activity over the training set from each pixel.
- No other preprocessing or data augmentation is applied at this stage; strategies are discussed elsewhere.

2.1 ImageNet Dataset Overview and Challenges

- ImageNet contains over 15 million labeled high-resolution images across roughly 22,000 categories.
- ILSVRC uses a subset with about 1,000 images per category, totaling 1.2 million training images.
- Images are collected from the web and labeled via crowd-sourcing, e.g., Amazon Mechanical Turk.
- Challenge: large-scale datasets require models with high capacity and techniques to prevent overfitting.
- Images are resized to 256×256 by rescaling and cropping, then mean activity is subtracted.

2.2 Data Augmentation Strategies

- Data augmentation involves label-preserving transformations like translations, reflections, and PCA-based color adjustments.
- These techniques effectively enlarge the dataset, reducing overfitting and improving generalization.
- For training, random 224×224 patches are extracted from 256×256 images, along with their horizontal reflections.
- At test time, predictions are averaged over five patches and their reflections to enhance accuracy.
- RGB intensities are also altered by adding scaled principal components, capturing illumination invariance.

2.3 Image Resampling and Normalization

- Images are down-sampled to a fixed resolution of 256×256 pixels.
- Rescaling ensures the shorter side is length 256, then crop the central 256×256 patch.
- No additional pre-processing is applied, except subtracting the training set mean activity.
- Training uses raw RGB pixel values, centered by mean subtraction, for input normalization.
- This simple resizing and centering prepare images for consistent input to the CNN.

2.4 Dataset Variants and Evaluation Protocols

- The ILSVRC-2012 test set labels are unavailable; performance is evaluated on validation set or via protocols.
- Our model's top-5 error rate on ILSVRC-2012 is 18.2%. Averaging predictions from multiple CNNs improves accuracy to 16.4%.
- Training on the full ImageNet dataset with pre-training and fine-tuning yields error rates around 15.3%.
- Error rates are reported as top-1 and top-5 on validation and test sets, with test labels often not publicly available.

3. Overfitting Prevention and Regularization

- Overfitting is a significant challenge in large neural networks trained on massive datasets.
- Techniques like data augmentation and dropout are among the methods used to prevent overfitting, as noted in the source.
- Data augmentation artificially enlarges the training set via label-preserving transformations, such as translations and reflections.
- Dropout involves randomly setting neuron outputs to zero during training, promoting more robust feature learning.
- These methods help improve generalization, but are not the only techniques available for regularization.

3.1 Data Augmentation for Overfitting Reduction

- Data augmentation artificially enlarges the training set with label-preserving transformations.
- Includes image translations and horizontal reflections to increase variability.
- Transforms generate many interdependent training examples, reducing overfitting.
- RGB intensity alterations via PCA add natural illumination variation, boosting robustness.

3.2 Dropout Technique and Its Application

- Dropout randomly sets neuron outputs to zero during training with probability 0.5, applied in fully-connected layers.
- This prevents neurons from co-adapting, encouraging more robust feature learning.
- At test time, all neurons are used, with outputs scaled by 0.5 to approximate ensemble averaging.
- Dropout roughly doubles training time but significantly reduces overfitting in large neural networks.

3.3 Additional Regularization Methods

- Weight decay with coefficient 0.0005 helps prevent overfitting.
- Early stopping based on validation error halts training before overfitting occurs.
- Response normalization layers create competition among neuron outputs, aiding generalization.
- Dropout randomly zeroes neuron outputs in fully-connected layers, reducing co-adaptation.

3.4 Effectiveness and Validation Results

- Dropout in fully-connected layers roughly doubles training time but significantly reduces overfitting.
- Data augmentation via image translation, reflection, and PCA color shifts enhances generalization.
- Response normalization layers following convolutional layers improve top-1 error by about 1.4%.
- Overlapping pooling ($z=3$, $s=2$) reduces top-1 error by 0.4% compared to non-overlapping pooling.
- Validation results show each regularization method contributes to improved model robustness and accuracy.

4. Training Methodology and Results

- We trained large CNNs on ImageNet subsets, achieving record top-1 and top-5 error rates.
- Utilized GPU-optimized convolution implementation and multi-GPU parallelization to handle model size.
- Applied extensive data augmentation, including translation, reflection, and PCA-based color jittering, to reduce overfitting.
- Dropout was employed in fully-connected layers, significantly improving generalization and reducing overfitting.
- Ensembling multiple CNNs and pre-training on the entire ImageNet 2011 dataset further lowered the top-5 error to 15.3%, demonstrating combined benefits of model averaging and transfer learning.

4.1 Optimization Details and Hyperparameters

- Weights initialized from zero-mean Gaussian distribution with std 0.01.
- Neuron biases in convolutional and fully-connected layers initialized to 1 for faster learning.
- Learning rate started at 0.01, reduced three times when validation error plateaued.
- Trained for roughly 90 cycles over 1.2 million images, taking 5-6 days on two GPUs.
- Used stochastic gradient descent with batch size 128, momentum 0.9, weight decay 0.0005.

4.2 Multi-GPU Training Strategy

- Training spread across two GPUs to handle large models exceeding single GPU memory.
- Kernels in early layers distributed with partial connectivity, reducing communication overhead.
- Layer connections are optimized: some layers connect only within the same GPU, others across GPUs.
- Communication occurs only at certain layers, balancing computation and data transfer.
- Two-GPU scheme improves top-1 and top-5 error rates compared to single GPU training.

4.3 Quantitative Performance Metrics

Model	Top-1 (val)	Top-5 (val)	Top-5 (test)
SIFT + FVs [7]	—	—	26.2%
1 CNN	40.7%	18.2%	—
5 CNNs	38.1%	16.4%	16.4%
1 CNN*	39.0%	16.6%	—
7 CNNs*	36.7%	15.4%	15.3%

Table 2: Comparison of error rates on ILSVRC-2012 validation and test sets. In italics are best results achieved by others. Models with an asterisk* were “pre-trained” to classify the entire ImageNet 2011 Fall release. See Section 6 for details.

- Our CNN achieves a top-1 error rate of 37.5% and top-5 of 17.0% on ILSVRC-2010.
- On ILSVRC-2012, the model attains a top-5 error rate of 18.2%.
- Averaging predictions over five CNNs reduces top-5 error to 16.4%.

4.4 Qualitative Analysis and Visualizations

- Figure 3 shows learned convolutional kernels with frequency, orientation, and color features.
- Kernels exhibit specialization: color-agnostic on GPU 1, color-specific on GPU 2.
- Top-5 predictions on test images reveal reasonable labels even for off-center objects.
- Feature activation similarity helps identify visually similar images, despite pixel differences.
- Visualizations provide insights into learned features and network behavior beyond quantitative metrics.

Thank you for listening!